Upcycle Compass · Decision memo

Date: 24 Oct 2026 · engine 0.5.0 · dataset 2026-10-24.3

Batch and use
groundnut cake → beverage (domestic)

Verdict ⚑ TEST-FIRST hypothesis-grade P(GO) ≤ **32%** S

Why

- G1 Aflatoxin B1: AFB1 not measured by HPLC; prior P(fail) = 0.53 [published]. Test a composite of ≥ 10 sub-samples (sampling error often dominates assay error).
- G5 Food-grade provenance: feed-grade or unknown supply and open or unknown storage: run a micro panel; re-source if possible
- Key evidence is transferred from other materials: must be demonstrated on this one.
- Base protein is about 48% against 60% required: flour cannot reach it; a protein concentration (fractionation) step is needed.

Safety gates
G1 Aflatoxin B1: TEST-FIRST · G4 Solvent residue: PASS · G5 Food-grade provenance: TEST-FIRST · G6 Allergen: FLAG · G7 Regulatory: FLAG

Recommended route
extrude → ae_iep (isolate): protein **88.5%** 84%–93% S transferred, P(spec) **68%** S transferred

Cost vs imported pea isolate
₹192 ₹105–₹958 S transferred (lower bound) vs **₹490** ₹407–₹583 A (bcd_only_igst_credited_22)

Test to buy first
Proximate + Kjeldahl protein: **₹1,51,823** S transferred

Decision readiness
hypothesis-grade: ✕ All safety gates pass; ✕ Best chain P(spec) ≥ 0.8; ? Remaining uncertainty EVPI / |E_now| ≤ 0.15; ✕ ≥ 70% of the top 5 value drivers graded A/B

Key limitations

- Inputs are sampled independently (no correlations between inputs).
- Functionality fit (solubility, emulsification, gelation) is a qualitative flag, not a computed score.
- 9 fractions have no data and are sampled over their full 0-1 range (assumed, grade D), e.g. ae_iep: trypsin_inhibitor reduction, extrude: mass yield, extrude: protein retention. Ranges are wide on purpose.
- Costs are lower bounds: missing QA cost, capital, chemicals, energy, labour, overhead, water inputs are excluded.

Sources

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3. Team review 'From By-Product to Functional Food: Valorization of Cold-Pressed Oilseed Cakes' (final draft, unpublished), Tables 2, 3, 7, 8, 11
4. IndiaMART trader listing (Mahalakshmi Oils): groundnut cake and de-oiled rice bran prices · https://m.indiamart.com/mahalakshmi-oils
5. Tridge: pea protein (80-85%) export prices, China, 2023-24 · https://dir.tridge.com/prices/grass-pea-protein/CN
6. USD/INR quotes, Sep-Oct 2026 (banklive.net) · https://banklive.net/en/convert/usd-to-inr
7. IndiaMART and TradeIndia pea protein isolate B2B listings · https://www.indiamart.com/proddetail/pea-protein-isolate-23662128530.html